

Database System Engineering and Distributed Backend Development

WarehouseIQ

Smart Warehouse & Inventory Management System

DBMS Project

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TEAM NUMBER- 4

SECTION-9



Problem Statement

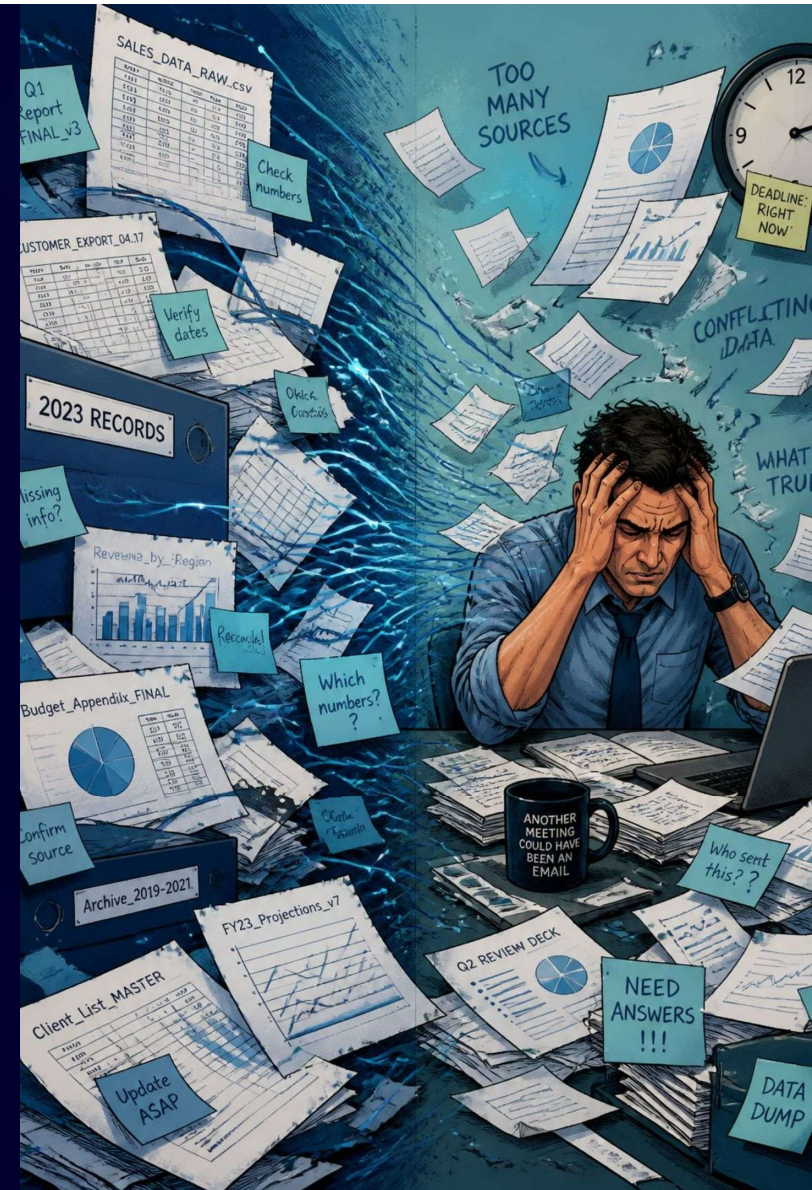
1 Traditional

Manual records · Scattered spreadsheets · Difficult stock tracking

2 Result

Stock inaccuracies · Missed low-stock items · Slow, unreliable reporting

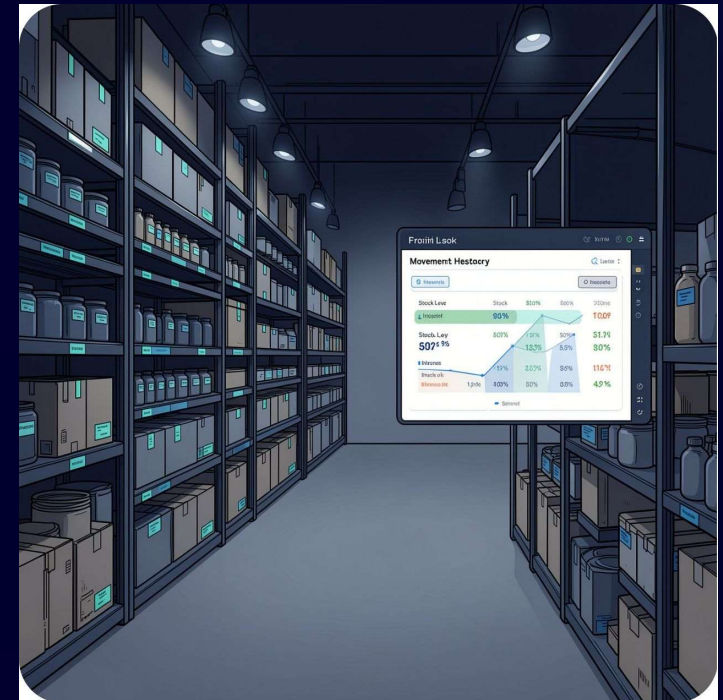
Fragmented data across people and files causes slow decisions and operational risk.



Everything your warehouse needs, in one place.

WarehouseIQ provides a centralized system for managing products, suppliers, warehouse information, inventory, and stock movements. It records stock entering and leaving the warehouse, maintains movement history, highlights low-stock products, and provides simple reports for better inventory visibility.

Products • Inventory • Suppliers • Stock Movements • Reports



Core Modules



01 Login

Secure, role-based access and single sign-on readiness.



02 Product Management

Canonical product records, SKU attributes and catalog control.



03 Supplier Management

Vendor profiles, lead times and contact history.



04 Warehouse Management

Location maps, bin assignments and capacity overview.



05 Inventory Management

Live stock counts, reserved vs available and valuation.



06 Stock In / Stock Out

Structured receiving and dispatch workflows with checks.



07 Movement History

Immutable movement logs for traceability and audits.



08 Dashboard & Reports

Concise operational dashboards and exportable reports.

Built With

An intentionally simple, maintainable stack:

Frontend

React + modern CSS for a responsive, accessible UI.

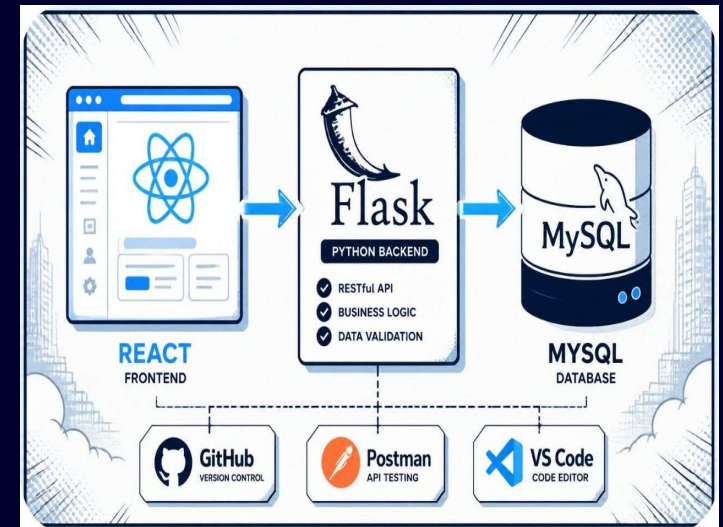
Backend

Flask (Python) serving a RESTful API and business logic.

Database

MySQL: normalized DB schema with transaction-safe operations.

Supporting tools: GitHub · Postman · VS Code





Thank You

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